PAVITHIRAN

Full Stack Engineer

Coimbatore, India | 8072004351 | pavithiranmailid@gmail.com

PROFESSIONAL SUMMERY

Senior Backend Engineer with 3+ years of experience in **FastAPI**, secure **API development**, real-time systems, and ERP System. Strong background in **PostgreSQL, Firebase, Docker, AWS ECS, CI/CD**. Domain experience in **Healthcare and ERP systems**.

WORK EXPERIENCE

Swar Technologies — **Coimbatore, India**

Full Stack Engineer & Software Auditor — **April 2025 - Present**

Full Stack Engineer

- Lead end-to-end web and mobile app development using **FastAPI, Flask, Firebase, SQL, Flutter, Bootstrap, and JavaScript**, delivering secure and scalable business solutions.
- Implemented a **modular, clean architecture** across all projects — significantly **reducing project initiation time** and improving code maintainability.
- **Migrated legacy systems from Flask to FastAPI** and adopted **SPA architecture**, enhancing performance, scalability, and developer efficiency.
- Built and enforced **global middleware, JWT authentication, rate limiting, exception handling, and** standardized API response classes for consistency and security.
- Set up **Git/GitHub CI pipelines** to automate version control, code review, and deployment workflows.
- Developed and deployed **static and dynamic web applications** across **Hostinger, Render, Vercel, and PythonAnywhere**, and packaged software into desktop .exe applications.

Software Auditor

- Conduct **end-to-end software audits** for clients **Sree Kumaran Thangamaligai(SKTM),** ensuring adherence to **SDLC practices**, code quality, and system performance standards.
- Audited and **prepared two major enterprise products — P2P (Purchase to Payment) and HR Payroll — for sales readiness**, ensuring code quality, stability, and documentation compliance.
- **Collaborated across multiple departments** (Purchase, Accounts, Billing, etc.) to identify process gaps and implement software enhancements, resulting in **significant man-hour reduction** and improved workflow efficiency.
- **Audited and supervised the KTM Web Billing System**, performing on-site visits across six **branch locations** to guide and support successful software implementation.
- Developed automated **testing and reporting systems** using Pytest and PyWinAuto, and implemented a containerized **server monitoring and alerting platform** with Prometheus, Grafana, and Alertmanager for 11+ servers.

Genpact India — **Hyderabad, India**

Technical Associate — **Dec 2023 - April 2025**

- Completed **Java Full Stack Development** training and developed an **IT Project Management System** during the program to enhance internal project tracking and collaboration.
- Worked on the **WMSA Project** as a **Technical Associate**, managing and resolving **business and customer support tickets** .
- Collaborated with cross-functional teams to analyze issues, streamline workflows, and ensure timely resolution of system incidents.
- Attended regular review meetings to align with client expectations, document progress, and provide technical inputs for service improvement.

Alt X Electronic — **Coimbatore, India**

Embedded Engineer(Co-Founder) — **Jan 2023 - Dec 2023**

- Founded and led **Alt X Electronic**, specializing in the **design and manufacturing of step-down transformers** and the development of embedded and IoT-based systems.
- Designed and delivered **college and industrial IoT projects**, including **automatic water-level control systems, drones, battle bots, home automation, Li-Fi communication systems, DTMF-controlled cars, air pollution monitoring, fertigation systems, and cold shed natural cooling and ventilation**.
- Conducted **technical training and internships** for engineering students on **PLC programming, SCADA design, and Industrial IoT (IIoT)** system development.
- Oversaw **end-to-end project design, prototyping, and testing,** ensuring reliable, cost-effective, and production-ready embedded solutions.

EDUCATION

- **Minor In AI** Sept 2024 -25
  Indian Institutes of Technology (IIT), Ropar
- **Bachelor of Engineering - Electronics and Communication** May 2019 - May 2023
  SNS College Of Technology, Coimbatore **CGPA - 9.12**

SKILLS

- Programming Languages: **Python, Java, Dart, JavaScript, HTML, CSS**
- Frameworks & Libraries: **FastAPI, Flask, Flutter, React, Bootstrap, Tailwind CSS, SQLAlchemy ORM**
- Databases: **SQL, PostgreSQL, Firebase, Hive, MS SQL Server**
- Tools & Platforms: **uv, Git, GitHub, GitHub Actions, Docker, AWS Elastic Container Service(ECS), PostgreSQL, Vercel, Render, Hostinger**
- Core Expertise: **RestFull API Development**, **WebSocket Implementation, Full-Stack Web Application Development, Mobile App Development**, Software Auditing, SQL Debugging

PROJECTS

Swar Technologies

Technologies Used: **Python, FastAPI, Flask, Firebase, SQL, Flutter, React, Bootstrap, JavaScript, HTML, CSS, Git/GitHub, Docker, Vercel, Hostinger, AWS, Prometheus, Granfana**

- **Hospital Management & Appointment System (Client – Nithya Clinic)** - Developed full-stack web and mobile applications to manage patient records, appointments, and clinic operations, improving efficiency and data accuracy.
- **Hospital Management System (Client – Sree Subham Nursing Home)** - Built a comprehensive web solution for managing patient data, appointments, and hospital workflows. Performed website migration for improved performance.
- **PO Management System (Client – AUV Engineering) -** Created web for automated purchase order creation, tracking, item, supplier management and reporting, streamlining procurement processes.
- **PO Management System (Client – Sree Venkateshwara Textiles & Chemicals) -** Developed both web and mobile applications for purchase order management, enabling real-time tracking, reporting, and cross-department workflow integration.
- **Loan Management System (Client – Royal Finance) -** Designed and implemented a mobile app to manage loan applications, approval workflows, and repayment tracking, enhancing operational efficiency.
- **Admin Panel & Internal Tools (Internal) -** Developed internal dashboards and management tools to standardize operations, project tracking, and reporting processes.
- **Websites & Deployment (Various Clients) -** Deployed multiple static and dynamic websites using Hostinger, Render, Vercel, and PythonAnywhere; converted web applications into desktop .exe software for client use.
- **Automated Project Checking & Report Generation (Client - SKTM) –** Built an automated project validation and report generation system using Pytest and PyWinAuto to perform functional and UI-level checks to reduce manual testing effort.
- **Server Monitoring & Alerting Platform (Client - SKTM) –** Implemented a Dockerized monitoring and alerting stack using Prometheus, Grafana, and Alertmanager to monitor performance and availability across 11+ servers with proactive alerts.
- **Work-Force-360 (Internal - currently working)** – Designed the full project architecture, **developed backend APIs**, and **implemented WebSocket-based real-time communication** for connecting industry workers with project needs

Genpact

Technologies Used: **Java,Spring Boot, MySQL, ServiceNow**

- **WMSA Project(Client - GE Vernova)** – Managed and resolved **business and customer support tickets** using ServiceNow, ensuring timely issue resolution and minimal service disruption. Collaborated with cross-functional teams to troubleshoot and streamline workflows.
- **IT Project Management System** – Developed an internal project management application using Java and MySQL, enabling project tracking, task assignment, and reporting, improving team collaboration and operational efficiency during the Java Full Stack training program.

Alt X Electronics

Technologies Used: **Embedded C, Bluetooth Module, ESP32, Wi-Fi Module, PIC Microcontroller, Sensors**

- **Designed, tested, and manufactured step-down transformers,** delivering reliable and efficient power solutions for academic and industrial automation applications.
- **Air Pollution Monitoring System** – Designed and implemented an IoT-based air quality monitoring system selected for **TANSCST funding with a grant of ₹10,000.**
- **Cold-Shed Natural Cooling & Ventilation System** – Developed an eco-friendly cold storage solution for farmers' vegetable preservation; **patented innovation** for sustainable agri-tech.
- **Braille Digital Book** – Created an assistive technology device for visually impaired readers, showcased at a startup summit under the innovation category.